GEp-4 Reboot Feasibility Study

Data anchors, present assessment, and critical-path questions

Updated 11 November 2026

Working configuration: 10.7 GeV beam, 75 μA, 30 cm LH_2,
positive-polarity HMS proton arm, and NPS electron arm near 3 m.

What we set out to establish

Physics reach

Does the HMS–NPS configuration provide enough accepted elastic events at $Q^2 \simeq 11\ \mathrm{GeV}^2$ for a competitive form-factor measurement?

NPS response and backgrounds

Can a high NPS threshold retain elastic efficiency while controlling electron, photon, and charged-hadron backgrounds?

Trigger and DAQ

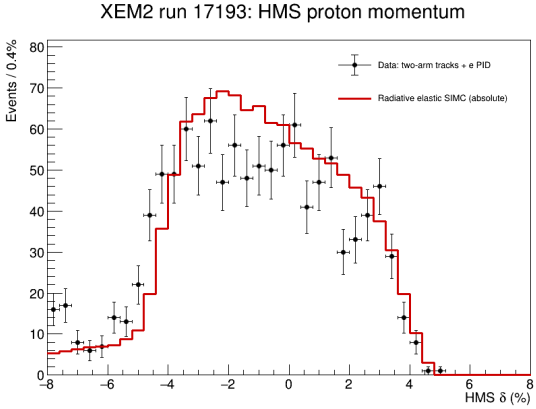
What are the real HMS and NPS singles rates, the accidental-coincidence rate, and the required Hall C DAQ capability?

Experimental design

Are the NPS distance, FPP configuration, HMS trigger logic, and coincidence timing mutually compatible?

Approach: anchor SIMC and GEANT4 to real Hall C data before extrapolating.

Elastic yield: a close HMS benchmark supports the proposal



XEM2 run 17193: HMS $+6.476$ GeV/c at 19.52°; proposed HMS $+6.735$ GeV/c at 18.80°.

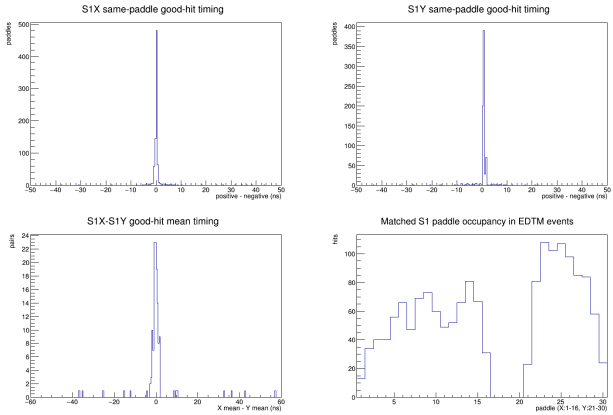
- Broad HMS acceptance: data/SIMC $= 0.936$.
- Tight elastic selection: $800/1034.34 = 0.7734$.
- At fixed 30 cm target length, NPS at 3 m gains a factor 4.11 over SHMS-like electron acceptance.
- Transfer from the XEM2 setting to $Q^2 \simeq 11\ \mathrm{GeV}^2$: SIMC ratio $= 0.8865$.
- Direct $Q^2 \simeq 11\ \mathrm{GeV}^2$ target scan: 20/10 cm $= 1.85$ and 30/10 cm $= 2.57$; the final 10 cm supplies 28% of the 30 cm yield.

45 PAC-day projection

Before NPS response	**10.69 M**
70% threshold + fiducial	**7.70 M**
80% threshold + fiducial	**7.51 M**

Response values remain provisional (3×3 energy, radiative photons, passive material). The sweeper will be absent in GEp; retaining it here is a small conservative effect.

HMS trigger: the practical S1 rate is 40–60 kHz



Run 17193 unbiased EDTM windows. A valid plane hit requires both PMTs on one physical paddle; valid X and Y paddle numbers are independent.

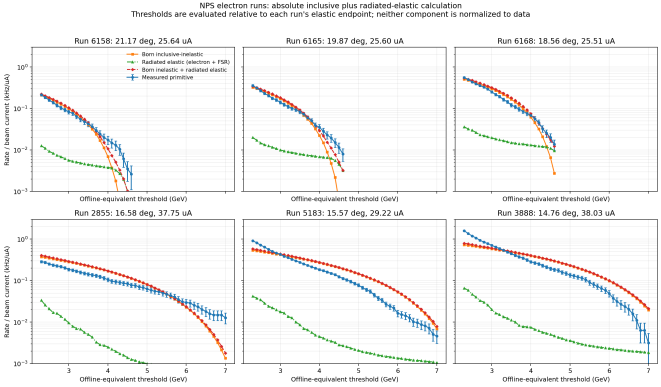
- The branch previously labeled "S1X AND S1Y" is actually **S1X OR S1Y**; its GEp-scaled rate is about 810 kHz.
- EDTM plane rates reproduce direct scalers to 3–4%.
- Requiring a double-ended paddle hit in both planes gives 12.73–14.84 kHz at the run conditions.
- Naive GEp scaling gives 51–60 kHz; scaler inclusion–exclusion gives 39.9 kHz.
- HMS 3-of-4 scales to about **2.9 kHz** if it is compatible with the FPP configuration.

Accidental scale

For 50 kHz HMS and 1 MHz NPS:

$$R_{\mathrm{acc}} = R_H R_N \Delta t = \begin{cases} 1\ \mathrm{kHz}, & 20\ \mathrm{ns} \\ 2\ \mathrm{kHz}, & 40\ \mathrm{ns}. \end{cases}$$

NPS singles: the inclusive-electron component is data anchored



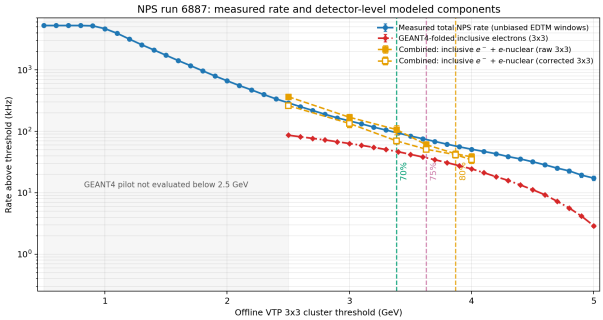
- Three-pass NPS electron data: generator/data $= 0.95$–1.35.
- Five-pass NPS electron data: trend reproduced; most discrepancies are conservative overpredictions.
- XEM2 SHMS electron-enriched trigger: 447.7 Hz predicted versus 592 Hz measured, or 74.8%.

GEp full-face prediction

Threshold	Inclusive e^-	+ radiated elastic
70%	14.34 kHz	**14.42 kHz**
75%	10.26 kHz	**10.30 kHz**
80%	7.05 kHz	**7.08 kHz**

These are genuine-electron rates, not the total NPS trigger rate.

Run 6887 establishes a large non-electron excess

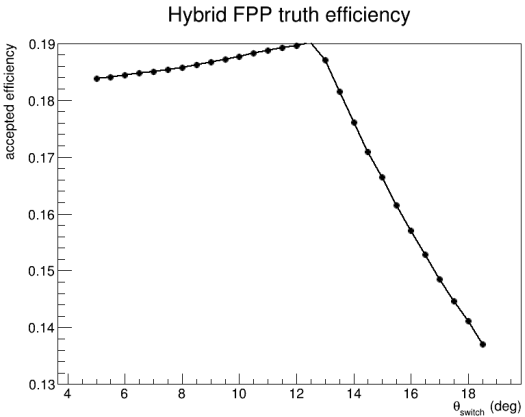


Run 6887: DVCS production, NPS on the photon arm; 8.456 GeV beam, 29.4 μA, 10 cm LH_2; NPS at 16.57° and 3.57 m. VTP single-cluster trigger/readout settings: 0.50/0.25 GeV; 8,972 unbiased 1-μs EDTM windows after the 25–35 μA current selection.

What the data establish

At 70–80%, inclusive electrons alone explain only 25–50% of the rate. Adding importance-sampled electronuclear production gives 107, 61, and 43 kHz with raw 3x3 energy, or 70, 51, and 42 kHz with the inherited correction. The six direct points use the 0.25-GeV production cut. **The background is now at the required scale**; the high-threshold deficit and 3x3 energy convention remain open.

A single analyzer preserves the small-angle FPP acceptance



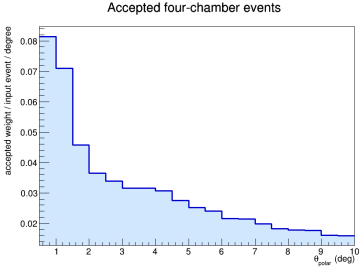
- Legacy FPP frame: one 22-inch CH_2 analyzer.
- $\theta_{\mathrm{polar}} < 12^\circ$: reconstruct with FPP1–FPP4; four chambers constrain the left/right ambiguity.
- $\theta_{\mathrm{polar}} > 12^\circ$: reconstruct with FPP1 and FPP2; the larger deflection resolves the ambiguity and avoids downstream cone-test loss.
- Truth efficiency peaks near the 12° transition.

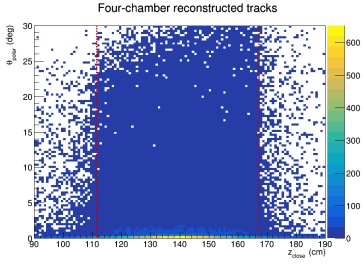
Four-chamber closure

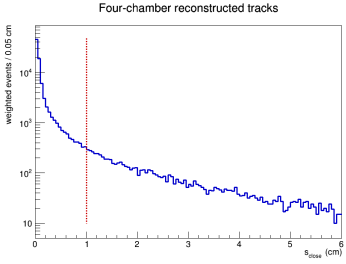
All 59,113 truth-selected events from 0° to 10° reconstruct in the ideal fit, with no large angular outliers.

Ideal closure: truth-associated hits and exact drifts; efficiency, resolution, noise, and pattern recognition remain.

Hall C chamber geometry and wire-level tracking







- Ported the four measured chamber positions and twelve U/X/V planes from the GEp-III parameters: 104/83/104 wires per chamber with 2 cm pitch.
- Digitize plane and wire number plus unsigned drift distance; enumerate the left/right signs and retain the minimum reduced-χ^2 $x(z), y(z)$ fit.
- Calculate the FPP observables relative to the HMS track and apply the cone test. The efficiency density at left includes the analyzer-volume, cone-test and 1 cm s_{close} requirements on truth-associated candidates.

Ideal closure: truth-associated hits and exact drifts; the fit receives neither the truth coordinate nor its sign. Detector response and multi-track recognition remain.

Single-track vs. Multi-track FOM

Use the same beam-polarization-weighted normalization per incident proton:

$$\mathrm{FOM} = P_e^2\,\epsilon_{\mathrm{FPP}}\langle A_y^2\rangle, \qquad P_e = 0.85.$$

Estimate / selection	Useful efficiency	A_y input	FOM / proton
Andrew's planning estimate	20% assumed	$\sim$0.050	$\sim 3.60 \times 10^{-4}$
Hall C: single tracks (no 1° cut)	12.492%	0.05859 RMS	3.10×10^{-4}
Hall C: smallest angle $> 1^\circ$	25.892%	0.06704 RMS	8.41×10^{-4}*

What explains the initial discrepancy

Our original 7.56×10^{-4} included multi-track events. Restricting to single tracks gives a result only **14% below Andrew's** 3.6×10^{-4} estimate, with the GEp-III A_y parametrization.

Andrew uses an estimated average A_y; the simulation integrates A_y^2 event by event. FOM agreement does not imply identical efficiency or cuts.

*Multi-track FOM is provisional: it assumes the single-track A_y model applies.

Analyzing-power parametrization and FOM weighting

The same construction appears in the SBS planning code and Andrew's description:

$$A_y(p, p_T) = \left(0.0342 + \frac{0.348}{p}\right) \sqrt{2eb}\, p_T e^{-bp_T^2}, \qquad b = 2.42614689.$$

b: inverse-variance-weighted mean of the three high-Q^2 fits; $A_y^{\max} = 0.0342 + 0.348/p$. Use $p_T = p \sin\theta$, with p, p_T in GeV/c.

Same exclusive single-track sample: $p = 6.7$ GeV/c, $\epsilon = 12.492\%$, $P_e = 0.85$.

Averaging convention	Effective A_y	FOM / incident proton
Square of mean: $P_e^2 \epsilon \langle A_y \rangle^2$	0.05484	2.714×10^{-4}
Mean square: $P_e^2 \epsilon \langle A_y^2 \rangle$	0.05859 RMS	3.099×10^{-4}

Event-by-event weighting gives 14.2% more FOM, not a factor of two. No parametrization mismatch was found.

Table VII's $\bar{A}_y$ is an asymmetry-calibrated effective average, not an RMS. Its $0.06 \le p_T \le 1.2$ GeV/c range also differs from our $0.07 \le p_T \le 1.0$ GeV/c selection (no explicit 1° cut here).

SBS asymmetry supports the analyzing-power scale

Andrew's preliminary $Q^2 = 11$ GeV^2 data give a sine amplitude $|s| = 0.02971 \pm 0.00403$. In the leading dipole approximation,

$$A_{y,\mathrm{eff}} \simeq \frac{|s|}{P_e P_\ell |\sin\chi|}, \qquad P_e \simeq 0.87.$$

Illustrative Born kinematics: $E = 10.7$ GeV, $Q^2 = 11$ GeV^2, $\mu_p G_E/G_M = 0.2$ give $\epsilon = 0.684$, $P_\ell = 0.729$, $P_t = -0.027$ for a fully polarized beam.

Estimate / assumption	Effective average A_y
SBS data, $\lvert\sin\chi\rvert = 1$	0.0468 ± 0.0064
SBS data, $\chi = 1.3$ rad	0.0486 ± 0.0066
Our simulated single-track mean	0.05484

The estimates agree within roughly one statistical standard deviation. Varying $\mu_p G_E/G_M$ from 0 to 0.4 changes P_ℓ by less than 0.5%.

Errors shown propagate the plotted asymmetry uncertainty only. This estimate uses approximate central kinematics and spin precession; acceptance averaging and background dilution are not determined here. It estimates a mean A_y, not the RMS, and does not validate multi-track FOM.

Source: A. Puckett, SBS Collaboration Meeting, 2 March 2026, slides 9, 10, 36.

SBS detailed yield accounting

Andrew's reconstruction study supplies the missing definition of the SBS "best case" sample:

$$\underbrace{12\ \mathrm{M}}_{\text{digi+recon, trigger, exclusivity, FT/FPP tracks}} \times \underbrace{0.70}_{\text{global performance}} \simeq \underbrace{9\ \mathrm{M}}_{\text{selected}} \times \underbrace{0.33}_{\text{useful polarimetry}} \simeq \underbrace{2.9\ \mathrm{M}}_{\text{useful events}} .$$

SBS estimate	Useful events	Projected $\sigma(\mu_p G_E/G_M)$
Andrew's stated quick-spreadsheet estimate	5.1 M	$\simeq 0.10$
Detailed reconstruction and cuts	2.9 M	$\simeq 0.13$

The uncertainty follows the expected count scaling: $0.10\sqrt{5.1/2.9} = 0.133$.

The 2.9 M count already includes the useful-polarimetry fraction; another 20% or 33% efficiency would double count it. The SBS generator is Born-only, so Andrew estimates that the useful yield remains 10–15% high.

Hall C and SBS FOM on a common basis

Using Andrew's $A_y = 0.05$ and $P_e = 0.85$, the detailed SBS sample gives

$$\mathrm{FOM}_{\mathrm{SBS}} = 2.9 \times 10^6 (0.85 \times 0.05)^2 = 5238.$$

Scenario	Total polarized FOM	$\sigma_{\mathrm{component}}$
SBS detailed, Born yield	5,238	0.0195
SBS with 10–15% radiative reduction	4,452–4,714	0.0212–0.0206
Hall C, exclusive single track	2,324–2,386	0.0293–0.0290
Hall C, smallest angle $> 1^\circ$	6,306–6,474	0.0178–0.0176*

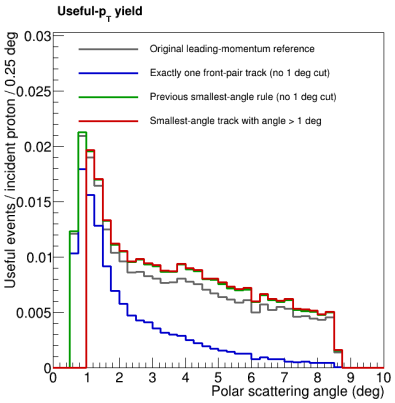
*The Hall C multi-track result remains provisional. It applies the GEp-III single-track A_y model to ideal truth-associated multi-track events.

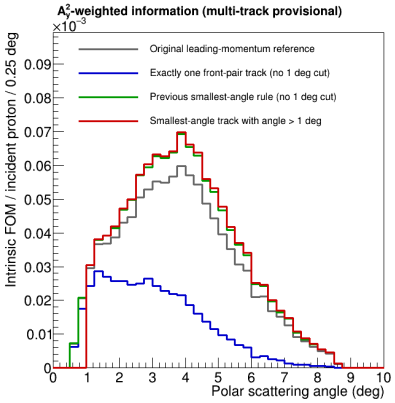
Resolution of question 3

The former quick SBS construction gave nearly the correct aggregate FOM, but Andrew's 2.9 M ledger supplies the defensible interpretation. Hall C becomes modestly stronger only if useful multi-track events survive realistic reconstruction and retain the assumed analyzing power.

Component errors are comparable feasibility metrics. A Hall C form-factor-ratio uncertainty still requires event-by-event spin transport and the polarization covariance.

Smallest-angle selection recovers useful multi-track events





Rule: choose the smallest polar angle strictly above 1°, then apply the existing chamber, cone, closest-approach, and useful-p_T cuts.

Result: 25,892 useful events, including 16,229 multi-track events, per 100,000 protons; 98.79% of the previous FOM retained.

Multi-track recovery could improve the FPP statistical reach

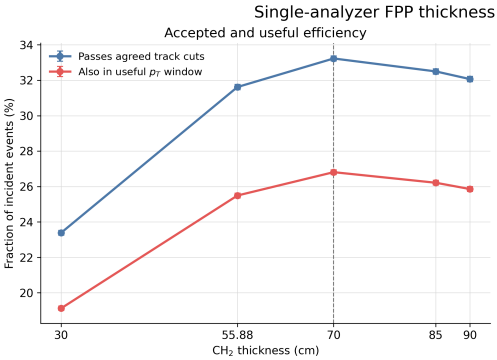
The single-track calculation supports Andrew's FOM scale. The simulation suggests that retaining some multi-track events could do better.

$$\sigma_{\text{component}} = \sqrt{\frac{2}{N\,\text{FOM}}}, \qquad N = 7.5\text{–}7.7 \times 10^{6}.$$

FPP selection	FOM / incident proton	$\sigma_{\text{component}}$
Exclusive single tracks	3.10×10^{-4}	0.0293–0.0290
Smallest angle $> 1^{\circ}$, including multi-track	8.41×10^{-4}	0.0178–0.0176*

*Provisional: GEp-III single-track A_y applied to ideal, truth-associated multi-track events. Detector efficiency, resolution, noise, and multi-track pattern recognition remain to be included. These are polarization-component uncertainties, not uncertainties in $\mu_p G_E / G_M$.

Selected-track yield per incident proton



Denominator for every point

100,000 unit-weight, 6.7 GeV/c protons incident on the analyzer.

Blue numerator: passes the agreed track cuts

- Among charged tracks crossing FPP1 and FPP2, retain angles strictly above 1° and choose the smallest angle.
- Require that chosen track to cross all four chambers and pass the all-four cone test, z_{close} inside CH_2, and $s_{\mathrm{close}} \leq 1$ cm.
- If it fails, do not replace it with another track. Outgoing momentum never enters the choice.

Red numerator: useful-p_T events

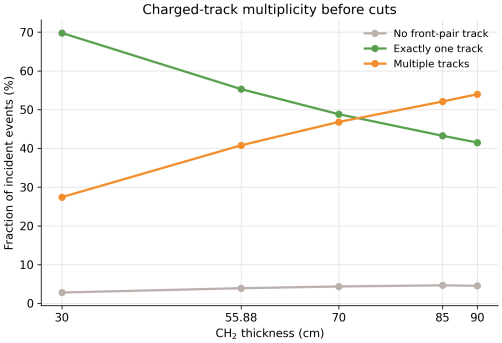
- Start with the blue events and require $0.07 \leq p_{\mathrm{inc}} \sin\theta \leq 1.00$ GeV/c, with $p_{\mathrm{inc}} = 6.7$ GeV/c.
- The selected track may be the primary proton or a charged secondary.

70 cm example: Blue: 33,229/100,000 = 33.229%; red: 26,810/100,000 = 26.810%.

The conditional ratio 26,810/33,229 = 80.68% is not plotted.

Charged-track multiplicity before analysis cuts

scan: event-level diagnostics



What is counted

Every charged candidate crossing both FPP1 and FPP2, before track ranking or any cone, closest-approach, angle, or p_T cut.

Each curve uses the same denominator of 100,000 incident protons.

Gray: no front-pair candidate.

Green: exactly one front-pair candidate.

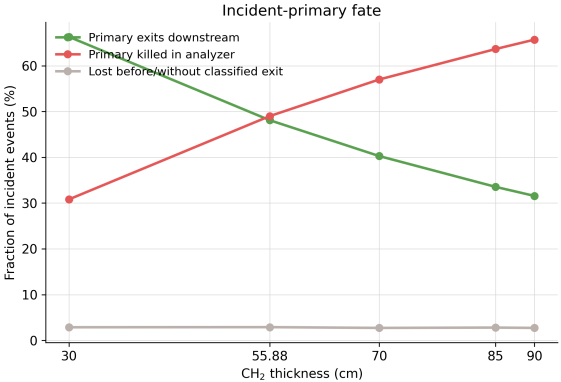
Orange: two or more front-pair candidates.

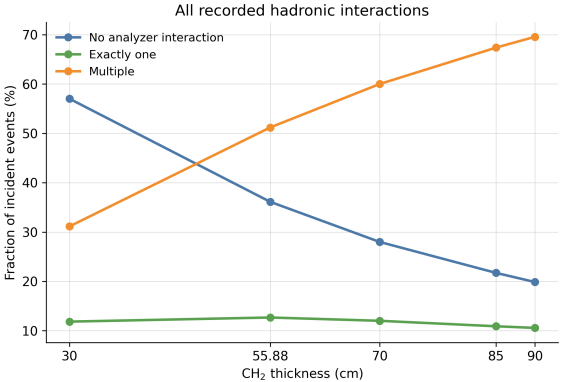
These three mutually exclusive categories sum to 100% at each thickness.

70 cm example

Gray: 4,358/100,000 = 4.358%. Green: 48,820/100,000 = 48.820%.
Orange: 46,822/100,000 = 46.822%. Total: 100,000 events.

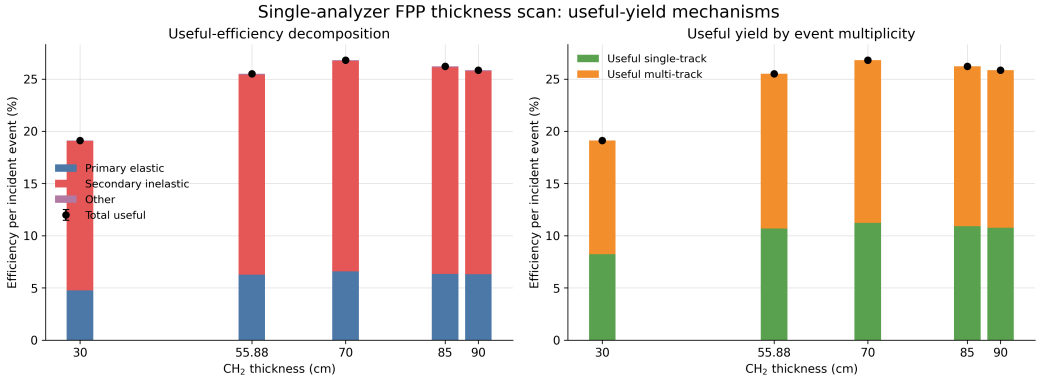
Primary loss grows rapidly with analyzer thickness





- The fraction of incident primaries killed in the analyzer grows from 30.8% at 30 cm to 65.7% at 90 cm.
- Events with multiple recorded hadronic interactions grow from 31.1% to 69.6% over the same range.
- Added material initially creates more useful scattering. Beyond about 70 cm, absorption and acceptance losses offset that gain.

Useful yield is dominated by secondary-inelastic tracks

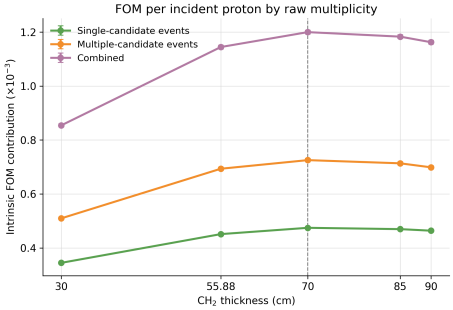


At the 70 cm maximum

The 26.81% useful efficiency contains 6.59% primary-elastic and 20.13% secondary-inelastic contributions. Raw single-candidate and multiple-candidate events contribute 11.21% and 15.60%, respectively.

The larger secondary-inelastic channel cannot inherit a single-proton analyzing-power model without dedicated validation.

Question 4: thickness-study assessment



- Single-candidate, multiple-candidate, and combined FOM all have a broad maximum near 70 cm. Extending to 85–90 cm provides no statistical gain in this scan.
- At 70 cm, the intrinsic contributions per incident proton are 4.745×10^{-4} from single-candidate events and 7.254×10^{-4} from multiple-candidate events.
- Their sum is 1.200×10^{-3}. The multiple-candidate term remains provisional because it uses the GEp-III single-track A_y model.

Current answer

Mechanically plausible, but 70 cm is the next design point. Final choice requires channel-specific polarimetry and realistic reconstruction.